

Algebra 2 *Advanced* - Unit 6 Bellwork: Week 2

Name: _____ Date: _____ Period: _____

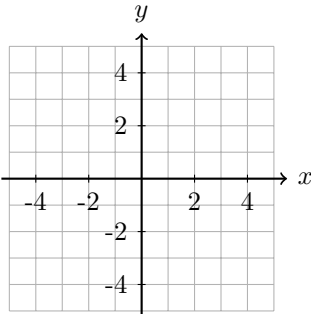
Day 5: Prior Knowledge for Graphing

1. Why does $f(x) = \sqrt{x}$ have a restricted domain but $g(x) = \sqrt[3]{x}$ does not?	2. If $f(x) = x^3$, find $f(-2) + f(2)$. $f(-2) + f(2) =$ What does this tell you about symmetry?
3. [SAT] Lines p and q are parallel. Line p passes through $(0, 3)$ and $(4, 7)$. Line q passes through $(2, -1)$. Find the equation of line q. A) $y = x - 3$ B) $y = x + 3$ C) $y = -x + 1$ D) $y = -x - 3$ Answer:	4. Solve: $2(x - 3)^{1/3} = (x + 5)^{1/3}$ $x =$

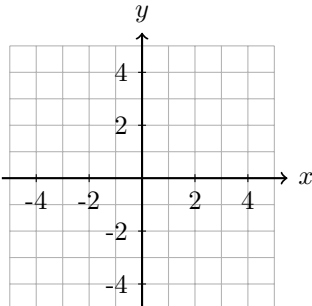
Day 6: Review - Parent Functions

1. Complete the table for $f(x) = x^3$: <table border="1"> <tr> <td>x</td> <td>-2</td> <td>-1</td> <td>0</td> <td>1</td> <td>2</td> </tr> <tr> <td>$f(x)$</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> Why does this show point symmetry about the origin?	x	-2	-1	0	1	2	$f(x)$						2. For $g(x) = \sqrt[3]{x}$, find $g(8) - g(-8)$. $g(8) - g(-8) =$ 3. If $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$, find: $f(g(64)) =$ $g(f(-3)) =$
x	-2	-1	0	1	2								
$f(x)$													
4. [SAT] The system $y > 2x - 1$ and $y < -x + 5$ is graphed. In which quadrant(s) does the solution region lie entirely? A) I only B) I and II C) II only D) I, II, III Answer:													

Day 7: Review - Cube Root Transformations

1. Write the equation for $\sqrt[3]{x}$ after: vertical stretch by 2, reflect over x-axis, left 3, up 4. $g(x) =$	2. $(8, 2)$ is on $f(x) = \sqrt[3]{x}$. Find the corresponding point on $g(x) = -\sqrt[3]{x - 1} + 5$. Point:
3. Graph $f(x) = \sqrt[3]{2(x + 1)} - 3$  Transformations: Center:	4. [SAT] Line m contains $(a, 0)$ and $(0, b)$ where $ab \neq 0$. If line n is perpendicular to m and passes through the origin, what is the slope of n? A) $\frac{a}{b}$ B) $-\frac{a}{b}$ C) $\frac{b}{a}$ D) $-\frac{b}{a}$ Answer:

Day 8: Review - Cubic Transformations

1. Describe all transformations and identify the inflection point: $g(x) = -\frac{1}{2}(x-3)^3 + 4$ Transformations: Inflection Point:	2. Graph $f(x) = 2(x+1)^3 - 4$  Transformations: Inflection:
3. A cubic has inflection point $(-2, 5)$ and passes through $(-1, 3)$. Write the equation in form $y = a(x-h)^3 + k$. $y =$	4. [SAT] If $2x - 3 < 7$, which is equivalent? A) $x < 5$ B) $x > -2$ C) $-2 < x < 5$ D) $x < -2$ or $x > 5$ Answer:

Day 9: Review - Writing Equations from Graphs

1. A cube root function passes through $(0, -2)$ and $(8, 0)$. The parent passes through $(0, 0)$ and $(8, 2)$. Write the equation. $f(x) =$	2. A cubic has inflection point $(1, -3)$ and passes through $(2, -1)$. Write the equation. $f(x) =$
3. Match each equation to its description: Reflected over y-axis A) $f(x) = \sqrt[3]{-x}$ Reflected over x-axis B) $f(x) = -\sqrt[3]{x}$ Horiz. compress by $\frac{1}{2}$ C) $f(x) = \sqrt[3]{2x}$	4. [SAT] $f(x) = 3x - 7$. If $f(a) = 5$ and $f(b) = 17$, what is $\frac{b-a}{2}$? A) 2 B) 4 C) 6 D) 8 Answer: